2 Importance of Civil Engineering

UNIT SPECIFICS

Through this unit we have discussed the following aspects:

- ***The importance of Civil Engineering in shaping and impacting the world***
    - *Responsibilities and Role in reaching SDGs*
    - *Direct and indirect impact with respect to SDGs*
- ***The ancient Marvels and modern Wonders in the field of Civil Engineering***
    - *Ancient Civilizations [4000 – 900 BCE]*
    - *Classical Period [1000 BCE – 1000 CE]*
    - *Renaissance & Age of Enlightenment [1400 – 1750 CE]*
    - *Modernism & Industrial Era [1750 - 1950 CE]*
    - *Contemporary style & Digital Era [1950 - present]*
- ***Future Vision for Civil Engineering***

Besides giving a large number of multiple choice questions as well as questions of short and long answer types marked in two categories following lower and higher order of Bloom's taxonomy, a list of references and suggested readings are given in the unit so that one can go through them for practice.

There is a "Know More" section, which has been carefully designed so that the supplementary information provided in this part becomes beneficial for the users of the book. It is important to note that for getting more information on various topics of interest some QR codes have been provided which can be scanned for relevant supportive knowledge This section mainly highlights applications of the subject matter for our day-to-day real life or/and industrial applications on variety of aspects, case study related to environmental, sustainability, social and ethical issues whichever applicable, and finally inquisitiveness and curiosity topics of the unit.

RATIONALE

This unit establishes the importance of the profession and practice of Civil engineering in the context of Sustainable development and future trends and needs. It further emphasizes the roles and responsibility of the civil engineer.

UNIT OUTCOMES

List of outcomes of this unit is as follows:

U2-O1: Knowledge on the role, impact/relevance, and importance of Civil Engineering

U2-O2: Knowledge on the outstanding Civil Engineering feats of ancient and modern times

U2-O3: Knowledge on the future trends in the field

Unit-2 Outcomes	EXPECTED MAPPING WITH COURSE OUTCOMES (1- Weak Correlation; 2- Medium correlation; 3- Strong Correlation)						
	CO-1	CO-2	CO-3	CO-4	CO-5	CO-6	CO-7
U2-O1	3		2	3	1	1	3
U2-O2	3	2	2	1	2	1	
U2-O3	2		1	2			3

The Civil Engineering discipline encompasses various specialisations, like, Construction Engineering, Structural Engineering, Geotechnical Engineering, Transportation Engineering, Environmental Engineering, Materials Engineering, Marine Engineering, Irrigation Engineering, Highway Engineering, Bridge Engineering, Hydraulic Engineering, etc. The built-environment, comprising of buildings, bridges, railways and roads, sewers and dams, power pants and transmission towers and lines, tunnels, canals, and waterways, are all results of these sub-domains. As discussed earlier, the built-environment, along with the inhabitants and social ecosystem it supports, has profound impact on the world and is a critical determinant of sustainable development. Hence, not only is civil engineering of importance for present needs of shelter, sanitation, transportation, and connectivity, but plays a significant role in addressing the issues and concerns arising due to the unmitigated urbanisation and economic development.

In the following Sections, a brief discussion on the importance of the discipline in shaping and impacting the world is followed by tracing the evolution and development of civil engineering practice through examples across time – from ancient era to present society. These ancient Marvels and modern Wonders reflect the society in which it thrived, with respect to the events and context discussed earlier in Unit 1, and paves way for the innovations of the future.

2.1 CIVIL ENGINEERING: SHAPING AND IMPACTING THE WORLD

The civil engineering market is expected to grow 25% each year (2022) up to 11.7 trillion US Dollars by 2025 (Global Market Insights, Inc.) due to the ever-increasing demand for infrastructure. In turn, increasing the need of competent engineers possessing sound knowledge of engineering fundamentals, and software and technical proficiency. In addition, a Civil engineer must hone skills as identified as the **Top Skills for 2025**, such critical thinking and creative problem-solving skills, the ability to communicate and collaborate effectively, coupled with strong leadership required for project management (World Economic Forum). Beyond the core activities of designing, i.e., conceptualise novel solutions, develop layouts and perform design calculations, and On-site supervision, civil engineers are **responsible** for;

- Oversee or perform soil tests, surveying operations, materials test, etc., and Analyse survey reports, maps, and other data to plan projects.
- Design and communicate ideas upon interacting with all stakeholders and identify possible design improvements, as well as incorporate Building codes, standards and guidelines into the design as required.
- Develop Project scope and timeline, and manage and monitor each stage of project,
- Assess environmental impact and risks, and ensure job site meets all legal guidelines and health and safety rules.
- Assist with staging, testing, and shipping of equipment prior to deployment.

- Prepare and present technical reports, analysis reports, cost estimates, Bill of Materials (BOM) and Bill of Quantities (BOQ), environmental impact statements.
- Submit permit applications to local or national agencies, and mitigate conflict.

Civil Engineering plays a **significant role** in offering methodologies, approaches, assessments of risks and impacts, as well as technologies and tools to assist decision-makers and technicians to achieve the Sustainable Development Goals (SDGs), in the following ways;

(i) ***Planning measures to mitigate and adapt*** to climate change, extreme weather events, earthquakes, droughts, floods, and other natural disasters.

(ii) ***Developing efficient and sustainable strategies*** for resource utilization while minimizing environmental impact and addressing unequal distributions.

(iii) ***Enhancing the safety of structures and infrastructures*** against exceptional loads and deterioration over their lifecycle.

(iv) ***Implementing a comprehensive risk management*** approach and appropriate technologies to reduce pollution and environmental degradation, thereby reducing vulnerability.

(v) ***Establishing safe drinking water and sanitation systems*** to safeguard human health.

A huge responsibility of achieving the SDGs lies on the shoulders of Civil engineers and the need for creative and sustainable solutions is the need of the hour. Most of the 17 SDGs are in some way connected to the discipline of Civil engineering, directly or indirectly, as illustrated below, highlighting the importance of the discipline.

- The **direct impact** that Civil engineering interventions can have, are on *SDGs 6 – Clean water and sanitation, 7 – affordable and clean energy, 9 – Industry, innovation and infrastructure, 11 – sustainable cities and communities, and 12 – Responsible consumption and production*.
- Under *SDG 1 – Eradicate Poverty*, its associated Target 1.4 focuses on "*access to basic services, ownership and control over land and other forms of property, inheritance, natural resources*", and civil engineers play an active role in planning land use.
- Under *SDG 2 – Zero Hunger*, Target 2.3 outlines "*By 2030, double the agricultural productivity*" and Target 2.4 stresses on "*By 2030, ensure sustainable food production systems and implement resilient agricultural practices*", which requires civil engineering interventions in form of irrigation, water and waste management, transportation, etc.
- SDGs *3 – Health and Well-being, 4 – Quality Education, 8 – Decent work and economic growth, 13 – Climate Action, 14- Life below water, and 15 – Life on Land* can all be improved by Civil engineering, through infrastructural solutions with reduced environmental impacts that are socio-economically respondent.
- Further, civil engineering projects and practice can become a potent medium to create awareness and statements, inspire change and imbibe best practices in the society towards *SDGs 5 - Gender Inequality, 6 – Reduced Inequalities, 16 – Peace, justice and strong institutions*.

2.2 ANCIENT WONDERS AND MODERN MARVELS

The prowess of a civilisation can be contemplated through the tangible and intangible cultural heritage, and the feats of design, engineering and construction of the ancient world indicates the advancement and creativity of these long-gone people. While events and aspirations dictated the progress of the civil engineering practice, the ancient wonders and modern marvels showcase the importance and impact of the discipline on mankind.

2.2.1 Ancient Civilizations [4000 – 900 BCE]

One of the first, most formidable, examples of civil engineering and urban planning maybe noted in the Indus Valley, with the city of Harappa being built during the Bronze Age, in the 4th millennium BC. Sanitation and wastewater systems were uncovered dating back to 2550BCE in the cities of Mohenjo-daro, Harappa and Lothal, and well planned urban housing made of strong *leeves* or earthen walls, with private toilets and drainage, networks of reservoirs and canals for irrigation, granaries with air ducts raised on high platforms, and designated public baths (refer Fig. 1), are some of the remnants of civil engineering of these flourishing cultures in the valley.

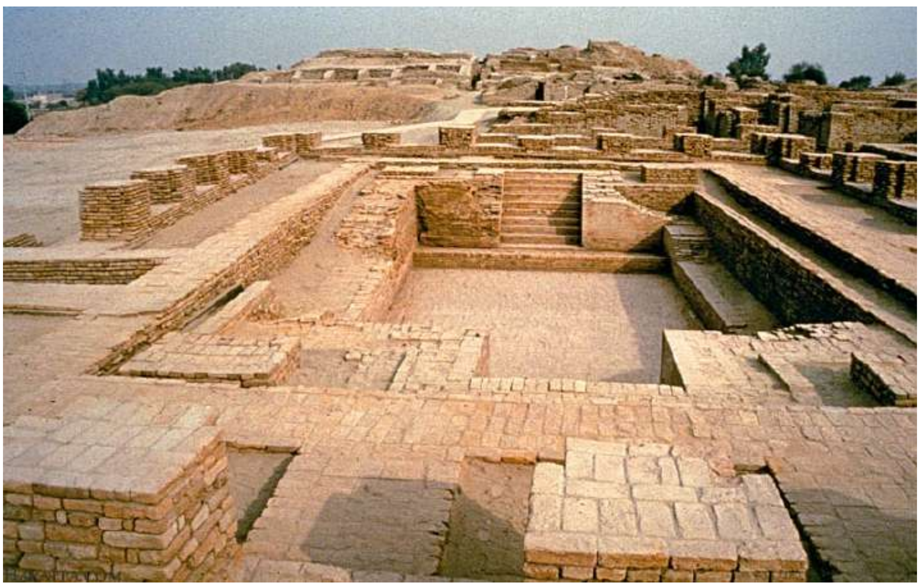

Fig. 2.1 : Great Bath at Mohenjo-Daro
(source : https://www.harappa.com/sites/default/files/slides/bath-indus.jpg)

In parallel, in ancient Mesopotamia (Iraq) along the rivers of Tigris and Euphratyes, the stepped pyramidal structure referred as **Ziggurat, at Ur** dates to 4000 BCE, with the 'White Temple' on top of it being built in circa 3500 BCE. Underwater channel systems, called '*qanats*', were constructed to route water from aquifers and wells to the surface for consumption and irrigation.

Civil engineering and architecture were at a peak in ancient Egypt. The **stepped pyramid** dedicated to King Djoser, of the 3rd Dynasty, located at Saqqara Necropolis in Egypt, was erected around 2700 BCE. Designed and build by architect-turned-God Imhotep, documents reveal his profound contribution of using shaped stones to create mammoth monuments with simple tools and mathematics. Shortly after, between 2575–c. 2465 BCE, Pharaoh Khufu of the 4th Dynasty, commissioned the building of the **three principal pyramids of Giza -** Khufu or the Great Pyramid (originally 481 feet); Khafre (471 feet); and Menkaure (213 feet), on the west bank of the Nile. These two sites, along with the ancient ruins of Memphis, are collectively recognised as a UNESCO World Heritage site.

Whilst in Europe, the **Stonehenge**, United Kingdom, was being completed approximately around 2,400 BCE. It comprises of two rings of standing stones with horizontal lintels on connecting them, arranged in a ditch with earthen embankments, with a trilithon – a structure of two large vertical stones with a horizontal resting across them, at the centre. The outer ring has vertical sarsen standing stone measuring 13 feet high, 7 feet wide and weighing roughly 25 tonnes, and the inner ring has smaller blue stones. The Stonehenge in Amesbury, Silbury Hill – a manmade earthen mound with pits and tunnels, and other henges at Avesbury, all are recognised as **UNESCO World Heritage sites**.

2.2.2 Classical Period [1000 BCE – 1000 CE]

What followed was an era of superlative design and construction, characterised by their architectural styles, hand in hand with the rise and fall of empires. From 1st millennium BC to 1st millennium AD, the Greco-Roman style dominated and was termed as 'Classical'. The Greek, and later the Roman, primarily military-oriented, focussed on expansion of their empires and hence, heavily invested on development of infrastructure. **Cities** emerged, such as, the capital Rome in Italy; Persepolis in Greece; Marseille in France; ports at Nucratis and later Alexandria, named after Alexander the Great, in Egypt; and Antioch in modern day Turkiye, once the seat of both the Byzantime and Roman empire. The Roman's championed road building and the first known roadway, the **Appian Way** or the 'queen of the roads', was constructed in 312BCE, connecting Rome with its allies in Capua. The Romans were also the first civilization of built permanent bridges or *Ponte,* as it not only played a strategic role in connecting the vast ends of the ever-growing empire traversing various rivers, but also was perfected in design to behave as aqueducts to carry water. **Pons Amelius**, the oldest stone bridge; Ponte Milvio, the second bridge; and Pons Fabricius, the oldest bridge still standing, all plied over the river Tiber, while the **Pont du Gard,** the tallest Roman bridge, carried water over 50km across southern France to the colony of Nimes.

Temples, urban settlements, such as cities and ports, and Universities were constructed, offering a broader picture of the priorities and wealth of the empires of the time. However, it is interesting to note that there was a strong underlying religious belief system in the Greco-Roman Pantheon and was a centric theme of construction. The **Temple of Artemis** at Ephesus, the **Statue of Zeus** at Olympia, **the Collosus** of Rhodes, **the Pharos** (lighthouse) at Alexandria, and the **Mausoleums at Halicarnassus**, all built at the time, are recognised as amongst the Seven Wonders of the Ancient World (other two being, the Great Pyramids of Giza and the Hanging Gardens of Babylon). Other notable religious structures built at the time are, the **Parthenon** dedicated to Goddess Athena at the Acropolis, a dedicated site atop Athens; **Temple of Jupiter Optimus Maximus** in Rome and those by Herod in the Judea, modern day Jerusalem.

In parallel, between the 10^{th} - 5^{th} century BCE, one of the first universities of the world was established at **Takshashila**, near the bank of the Indus River, India. It was a centre of great learning under the Indo-Greeks, with many scholars from all over the world. Chanakya, the noted scholar and Prime Minister of emperor Chadragupta Maurya, was a key figure at the university.

Further north, under the vision of the first emperor of united China, of the Qin dynasty, several existing piecemeal defensive wall structures were connected in the 3rd century BCE to develop a singular system of fortification, consisting of river dikes, bulwarks, and natural terrain, extending form the eastern Hebei province to the Gulf of Chihli. The early **'Great Wall'** was built with rammed earth, stones and wood, and later fortified with bricks and tiles with lime mortar, having passageways blocked by wooden gates. The construction of the Wall flourished under the Ming dynasty and continued till 17^{th} century CE.

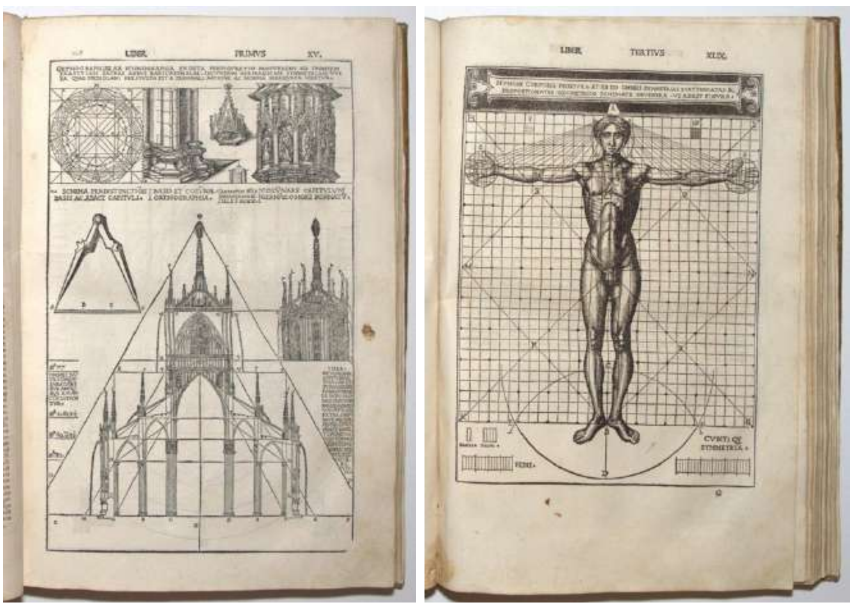

Fig. 2.2 : Pages from De Architectura [1475-1543], first vernacular edition, translated by Cesare di Lorenzo Cesariano (source : https://www.bada.org/object/first-vernacular-edition-de-architectura-vitruvius)

The development and eventual documentation of the **Classical style** can be traced to the earliest temples in Greece, such as that at Samos, built in timber framing. With change in material, from timber to stone, the scale and proportions grew, whilst retaining and refining the design elements and principles, that are still seen prevalent today. This was later composed as a ten-book treatise named, '*De Architectura*' (refer *Fig. 2*), authored by architect and military engineer, Marcus Vitruvius Pollio around 15c. BC, under the patronage of Caesar Augustus. It covered building materials, construction techniques and machines, guidelines on design of temples, civil and domestic buildings, common facilities, such as, pavements, water supply, aqueducts, as well as discussed, scientific and aesthetic principles, such as, geometry, measurement, buoyancy (Archimedes' principle), astronomy for sundials, and '***Orders of architecture***'. In addition, the first book opened with discourse on Town planning, architecture, and civil engineering, and elucidated the qualifications of an architect-engineer. Vitruvius proposed the triad of '*utilitas, firmitas, venustas'* (utility, strength, and beauty), as aesthetic characteristics of a good design and further developed the terms, order, arrangement, fit and proportion, which inspired Leonardo da Vinci to later illustrate the '*Vitruvian Man*'.

The 1st millennium AD saw the loss of Antiquity, as it gave way to political ambitions and philosophical schools of thoughts, led by thinkers such as, Plato and Socrates, leading to a shift from building of religious centres and temples to that of political and social interests, such as, the **Colosseum in Rome** (refer *Fig. 3*) and various triumphal monuments, such as, **Arch of Titus** and **Trajan's column**. But in the second half of the millennium, civil infrastructure continued to be built, such as, **Alcantara bridge** and **baths of Caracalla**; the rise of Christianity once again piqued interest in religious buildings. While several existing Greco-Roman temples and grottoes were converted in Christian churches, the new Roman Christian emperors took upon themselves tasks to build several basilica across their kingdom, from the **Basilica of Maxentius and Catacomb of the Via Latina** in Rome, and the **Papal basilicas** in present day Vatican City; to the *Aula Palatina* or Basilica of Constantine at Trier, Germany, and the **Hagia Sophia** (today known as the Grand Mosque) in Istanbul, Turkiye.

In 5th century CE, another profound institute of education **- Nalanda University,** in present day Bihar, India, came into establishment under the vision of the Gupta empire; and monolithic, rock-carved artistry and construction thrived under the patronage of the Pallava Kingdom in the southern state of Tamil Nadu. The famed Shore Temple, the *rathas* or temple-chariots dedicated to the Pandavas, several *mandapas* or pavilions, and noteworthy rock reliefs, such as '*Arjuna's penance*', at **Malappuram** (or Mahabalipuram), has earned recognition as a UNESCO World Heritage Site. Meanwhile in China, the world's oldest open-spandrel segmental arch-bridge of stone, fondly called Ānjì, meaning 'safe crossing' or **Zhaozhou Qiáo,** was constructed in the Hebei Province between 595-605 CE. In the 8th century CE, the **Seokguram grotto**, as part of Bulguksa Temple complex, in South Korea, was built and is recognised as National treasure 24 and as a UNESCO World Heritage site.

Thence onwards, the story of civil engineering runs hand in hand with architecture, largely responding to the human condition and the social context, with respect to, the visual language and aesthetics, structural layouts and buildings elements, construction techniques and materials, and the artist-engineer's creativity. This led to the development of styles categorised into 'periods' or 'movements'.

Following the Classical period, and before the next grand period of the Renaissance, which translates to 'rebirth', where two shorter styles that were found across Europe, namely;

> **Romanesque**, found mostly across Medieval Europe between 1050-1170, retained several features of the Roman style and is characterized by thick, heavy piers, narrow windows, stained glass, semi-circular arches, and towers, as in the Tower of Pisa; and **Gothic**, or *Opus Francigenum*, meaning "French work", was prevalent in the Late Middle Ages in France, between 900-1300, could be distinguished by pointed arches, high vaulted ceilings, flying buttresses, and vibrant interiors with stained glass windows, gables, colourful tapestries, etc. Noteworthy examples are, the **Cathédrale Notre Dame** in Paris**,** France, a UNESCO world heritage site and the Milan Cathedral or ***Duomo di Milano***, Milan, Italy.

Fig. 2.3 : Different Architectural styles across Italy (*Left*) Colosseum, Rome, Italy (source : photographed by Author, 2015), (*Right*) Tower of Pisa, Italy

Fig. 2.4 : Examples of Gothic Architecture across Europe : (*Left*) A model of Notre Dame Church exhibited inside the premise, Paris, 2019; (*Right*) A picture of the Duomo di Milano, Milan, Italy, during restoration in 2015

(source: photographed by Author)

2.2.3 Renaissance & Age of Enlightenment [1400 – 1750 CE]

Renaissance, emerged as a response to the over-embellished previous styles drawing inspiration from the Classical in the early 15th century at Florence, Italy, and spread across rest of Italy, France, Germany, Russia, Spain and England. It was characterised by use of arched openings, hemispherical domes, vaulted ceilings, orderly arrangement of columns, and formal landscaped gardens, exemplifying proportion, symmetry, and order. Several artist-architect-engineers, such as, Bruneslleschi, Boticelli, Michelangelo, Alberti, Palladio, Bramante, Bruni and Leonardo da Vinci, and great men of science and philosophy, such as, Copernicus, Galileo, Francis Bacon, Newton, Keplar, Machiavelli and More, flourished in this period. This age of discovery paved the path to several scientific and technological advancements of today.

Filippo **Brunelleschi**, popularly known as the 'Father of Renaissance', designed the *Basilica di San Lorenzo*, housing the pulpit and tondo works by **Donatella**; and engineered the world's largest brick vaulted dome, measuring 45.5m width and 116m height, for the *Santa Maria del Fiore* or Florence Cathedral. Another important advent of the Early Renaissance was the first theoretical ten-book compilation 'on the Art of Building', *De re aedificatoria*, by Leon Battista **Alberti**. It is also the first printed book on architecture, dated 1485, followed by the first edition print of Vitruvius' *De Architectura* (refer Fig. 2).

In the following High Renaissance period, Bramante, under the patronage of the Duke of Milan, incorporated classical styles in contemporary buildings bringing about a unique character to this period. Two noted examples are, the abbey church of *Santa Maria delle Grazie*, where he introduced the choir and a hemispherical dome with arched classical openings, rooted within an octagonal drum. He also worked on the cathedral of Pavia, the cloister of San Pietro in Montorio, and the *Cortille del Belvedere*, Vatican. In 1506, his design of the rebuilding of St. Peter's Basilica was selected, however, owing to his death the new chief Architect and noted artist, **Michaelangelo**, stayed true to the original proposal of the Greek-cross plan by Bramante. Another master builder of the time, **Andrea Palladio**, titled the "most influential architect of the Renaissance", adopted the Classical elements in a less grandiose manner and developed from it unique features, such as, the Palladian Arch - a scaled triumphal arch with square-topped opening for windows, the symmetric domed hall with identical four facades with orders supporting triangular pediments, as in the Villa Capra, famously called 'La Rotonda'. He also authored the "*I Quattro Libri dell'Architettura*" or Four Books on Architecture, published in 1570.

During the Late Renaissance, also called **Mannerism**, the trend continued where classical elements were reinterpreted as is seen by Michelangelo in the **St. Peter's Basilica.** The cupola of the basilica was designed as two masonry shells, one sitting into another, held by ribs which supported a massive roof lantern, and the exterior boasted a 'giant order' (see Fig. 5), defining the external square, held together by a wide cornice – creating a timeless masterpiece.

Following Renaissance, two styles emerged – the **Baroque**, and later, more flamboyant, **Roccoco**, between 1600-1755 as a sharp contrast to Renaissance. With an attempt to apply certain classical elements to regional architecture, without regard for the fundamentals of composition characterising Classical, these two styles are characterised by geometrically exaggerated, ornate and colourful structures, such as the **Palace of Versailles in France**, and **St. Paul's Cathedral in London**, and were seen in conjunction with the Counter Reformation movement.

Fig. 2.5 : St. Peter's Square, Vatican, boasting Michelangelo's 'giant order'
(source : photograph by Author, 2015)

2.2.4 Modernism & Industrial Era [1750 - 1950 CE]

Civil engineering remained at the helm during the **First Industrial Revolution**, with industries springing up and connectivity, transportation and sanitation became a dire requirement. Several bridges, canals and sewers were commissioned to carry effluents, goods and people, to and fro from industries. Noteworthy designs include *Bridgewater canal*, built by engineer Brindley in 1761, having several tunnels allowing direct travel to coal mines; and the world's first cast-iron, arch bridge over the River Severn at Coalbrookedale in 1776, where Pritchard used the brittle metal to construct a semi-circular structure that transfers load onto abutments, while held in compression to counter cracks from propagating – a first of its kind.

In 1771, John Smeaton the first self-proclaimed civil engineer, formed the **Smeatonian Society of Civil Engineers**. He constructed the Eddystone Lighthouse, off Cornwall, England, and developed 'hydraulic lime' and techniques of dovetail joints and dowels for securing blocks.

Road maintenance also became a priority in the 18th century, with the establishment of turnpikes trusts and toll roads. **Design and construction details of roads** saw significant improvement, with new methods proposed by Pierre Tresaguet in 1764 and later, improved by his mentee, John Metcalf who proposed convex shaped surface to allow rainwater to drain. Thomas Telford suggested small stones on the rock foundation further covered with a mixture of stone and gravel, having elevated pavements on either sides, known as 'Telford pitching', and John McAdam developed the '**macadamisation**' where soil foundation is layed with aggregate layers of angular stones and gravel. This design was later patented as the 'Tar Mc Adam road' by civil engineer, Hooley 1901, as is used today.

Soon railways spread rapidly and took over as the preferred mode of heavy goods transportation, with engineer Stephenson designing the **first steam locomotive** in 1825, and later improving it under the *Liverpoool and Manchester railway* (L&MR) in 1830.

A prolific civil engineer of the time, **Isambard Brunel,** pushed the boundaries of engineering and expanded his role as engineer-architect and designer. By 1833, the *Great Western Railway* (GWR) was commissioned under the able guidance of who had assisted on the *Thames Tunnel* project which has been incorporated today as part of the London Underground. Brunel, also has to his credit, design of several remarkable bridges, with the *Cliffton Suspension bridge* over Avon spanning 702 m - the longest at the time, as well as for, the world's longest steamship of the time, '*Great Western'*, constructed of wood and reinforced with iron diagonals and bolts, with improved surface condenser incorporated. He continued his experiments and further designed the first modern ship, *Great Britain*, which was propeller-driven and iron-hulled. The *Great Eastern*, his third transatlantic ship design, was the most technologically-advanced and luxurious ship of the time and is famed as one of the seven wonders of the Industrial Revolution. However, it aided the laying of oceanic telegraph cables rather than plying passengers, and proved to be pivotal in connecting America and Europe. He also designed the '*Three bridges'*, London, that allowed the routes of Great Western and Brentford Railway, Grand Junction canal and Windmill Lane; and the Renkioi Hospital, a first of its kind hospital in 1845, comprised of pre-fabricated wood and canvas that could be shipped and assembled on-site, far away in Tukiye, where the Crimean War was underway.

During the **Second Industrial Revolution**, when the spotlight was on the new nation of United States of America (USA), the **Neoclassical style,** as the name suggests, a revival of Classical style, was adopted to communicate a grand and powerful aesthetic in Gilded Age. Characterised by triangular pediments, free-standing columns, balustraded balconies, pronounced cornices, and symmetry, the White House and overall, the Capitol complex in Washington DC, it is reminiscent of Palladian architecture. Neoclassicalism also spread to England, France and Russia, with some of its outstanding advocates being, Robert Adam, John Soane and Claude-Nicolas Ledoux.

Modern design as understood today, epitomized by metal construction**,** first sprouted its presence with the **first two exhibition pavilions of the World Fairs** or the "*Exposition Universelle*". The two were, the **Crystal Palace** in Hyde Park, London, a cast-iron and plate-glass structure housing 92,000 sqm exhibition space designed by Joseph Paxton in 1851; and the **Eiffel Tower**, Paris, locally referred to as "*La dame de fer*" (Iron Lady), a wrought-iron lattice tower designed by architect, Stephen Stauvestre, along with structural engineers, Maurice Koechlin and Emile Nouguier. With further industrialisation supporting mass production of steel and glass, these materials became vastly employed as they imbibed a sense of never-before appeal, and further motivated futuristic styles.

Fig. 2.6 : The Great Exhibition pavilions - (*Left*) Crystal Palace, London, (*Right*) Eiffel Tower, Paris

Fig. 2.7 : Examples of Art Nouveau (*Left*) Sagrada Familia, Barcelona, and Art Deco style (*Right*) Chrysler Building, Chicago

Between 1890-1910, aversive to historicism and enthusiastic of modern life, the unique style of **Art Nouveau** developed characterised by organic lines and sinuous forms, achievable by steel frames and exaggerated by glass panes. The style was called *Jugendstil* in Germany, *Sezessionstil* in Austria, *Stile Floreale* (or Stile Liberty) in Italy, and *Modernismo* (or Modernista) in Spain; and was employed in building design by architects, Louis Sullivan across Chicago and further accentuated by artist-architect, Antoni Gaudi in Spain. Sullivan advocated the development of original forms and ornamentation, and is identified as '**Father of skyscrapers'**, exemplified in his design of the Wainwright Building in Chicago. In his firm worked a young civil engineer and aspiring architect, Frank Llyod Wright, who later was called the '**Father of modern architecture'** and like his mentor, developed a style unique to America

– the Prairie style, illustrated through large cantilevers and use of glass held with metal, like the Robie house or 'Fallingwater' – the most famous house in the world.

It was followed by **Art Deco style**, also referred to as style modern, which exhibited an affinity with machines as a reflection of modernity. While a short-lived movement between the 1920's and 30's, it had an indelible mark on the skyline of the emerging nation of USA, with iconic structures, such as, the Empire State Building, Rockefeller Center and Chrysler Building. This style was characterised by stepped gables, sculptured panels, ornate geometry, and cubic forms, along with use of unorthodox materials like, exposed steel and aluminium, decorative glass, ceramic and even, stucco and terracotta. In contrast, Europe saw the development of the '**International style**' characterised by rectangular forms, cantilevered projections, flat roof, ribbon windows, curtain glass, asymmetric facades and lack of ornamentation, as practiced by Le Corbusier in France, and Walter Gropius and Mies van der Rohe in Germany.

Various other styles such as, *Futurism, Constructivism, Brutalism, De Stijl, and Bauhaus*, all encompassed under the term **'Modernism'**, prevalent during the early 20th century, were grounded on the principle of '*Form follows Function'* as stated by Louis Sullivan. Fuelled by the abundant availability of mass-produced modern materials; they were a stark deviation from the traditional styles, characterised by functionalism, lack of ornamentation, and rational use of modern materials true to their nature, with a keenness for structural innovation. Philip Johnson's Glass House is an exquisite example.

2.2.5 Contemporary style & Digital Era [1950 - present]

Post-World War II, the international style flourished in the US for commercial buildings, as evident in the Seagram Building designed by Mies van der Rohe, a champion of minimalism famous for his aphorism "*less is more*".

But soon, in the 1960's, there was an antagonism towards the bleakness of modernism, and newer trends, such as, the introduction of the principles of **Beaux Arts style** by Louis Kahn and Eero Saarinen; and the need to focus on placemaking, with the local conditions and contexts through incorporation of vernacular by Robert Venturi, who published '*Complexity and Contradiction in Architecture'*, gave rise to the **Post-Modernism.** It challenged its predecessor with asymmetry, ornamentation, historical details and familiar motifs, and was characterised by eclectic inspirations and kitsch aesthetics, with focus more on form over function. Some of the notable examples are, the Lotus Temple in New Delhi, the Sydney Opera House, and the Guggenheim Museum, Bilbao, by pioneer Frank O'Gehry.

The concept of modularity was also catching on, with Archigram collective proposing a mega-scale modular concept called Plug-in City in 1964 and Shafie Moshdie designing the Habitat 67, a modular experimental housing. In Japan, the '***Metabolism' style*** developed, focusing on modularity, flexibility and interchangeable units, as exemplified by the Nagakin Capsule Tower designed by Kisha Kurokawa in 1972.

Fig. 2.8 : Examples of Modernism - (*Left*) Frank Llyod Wright's *Fallingwater*, Pennsylvania, (*Right*) Philip Johnson's *The Glass House*, Connecticut

Fig. 2.9 : Examples of Contemporary Style - (*Top Left)* Frank O'Gehry's Guggenheim Museum, Bilbao,
(*Top Right*) Nagakin Capsule Tower Kisho Kurokawa's *Nagakin Capsule Tower*, Tokyo,
(*Bottom Left*) Mies van der Rohe's *Seagram Building*, New York City,
(*Bottom Right*) Renzo Piano and Richard Roger's *Centre de Pompiduo*, Paris

Meanwhile in India, between 1950's-70's, the young nation also felt the ripples of these movements, with the vision of setting up a model city – Chandigarh. The task of designing a modern yet culturally sensitive aesthetic, Le Corbusier devised the city's masterplan by applying the concept of *Unité d'Habitation*. With the Capitol Complex at its heart, clean geometry and concrete facades were interspersed with motifs and symbology, and compositions of colour. And in 1961, the year of completion of the construction of the Capital Complex – a UNESCO World Heritage site, another grand project was in motion. Louis Kahn was commissioned the tasks of designing the IIM Ahmedabad (1962-74), a vision of Dr. Vikram Sarabhai and Kasturbahi Lalbhai, to develop professionals for the country's growing industrial progress.

The introduction and widespread adoption of the use of CAD software between the 1960's – 1990's led to the widening of architectural styles beyond post-modernism to high-tech or **Structural expressionism**, which aimed at showcasing the underlying function-structure of the buildings, from exterior to the interior, through the use of advanced technology and materials. Aluminium, steel, glass and concrete offered a sense of grandiose and honesty, with open plans allowing reconfigurable spaces, large overhangs and lack of load bearing walls. The World Trade Centre (1971) in New York by Yamasaki, the Centre Pompidou (1977) in Paris by Renzo Piano and Richard Rogers, and the Burj al Arab (1991) by Tom Wright are some famous examples. Improved design tools furthered modular construction and led to the development of the **Klip House concept**, designed by Interloop between 1997-2001, where modules could be snapped together to build a unit. Presently, the world's tallest building, the Burj Khalifa (2010) with the height of 830m to the tip, housing 163 floors, is an epitome of contemporary architecture. The three-leafed structure is an abstraction of the desert flower, Hymenocallis, twirling into a spire as it gains height.

Hand in hand, another trend of the present times, propelled by the development of CAD and other design tools, is **Parametricism.** Characterised by the use of algorithmic equations and computational tools leveraged to generate varied and almost impossible forms which are complex yet fluid, can be traced back to 1997. Earlier famed architects, Frei Otto and Antoni Gaudi are often considered inspirations for this style, personified by *Pritzker Awardee, Zaha Hadid*. The Guangzhou Opera House, at Galaxy Soho in Beijing China, and the Hyder Aliyev Centre in Baku, Azerbaijan, are a few of her exemplary works.

While using parametric design not only enables designers to optimise the planning and improve efficiency of the design, attempts to incorporate sustainability factors is currently being pursued by noted firms, such as, Ai Space Factory, Foster & Partners, etc. The OPPO R&D headquarters, named the **'Infinity Loop',** *designed by Bjarke Ingels Group (BIG)*, China, boasts an elegant form that is self-shaded, thereby, it reduces the energy usage and increases natural light. Another remarkable example is the '*Heart of Yong'an*' by TJAD, China, where algorithmic modelling has been used to create a hyperbolic, single slope, curving roof structure, emulating the surrounding mountains, with rammed earth walls and local blue tile roofing.

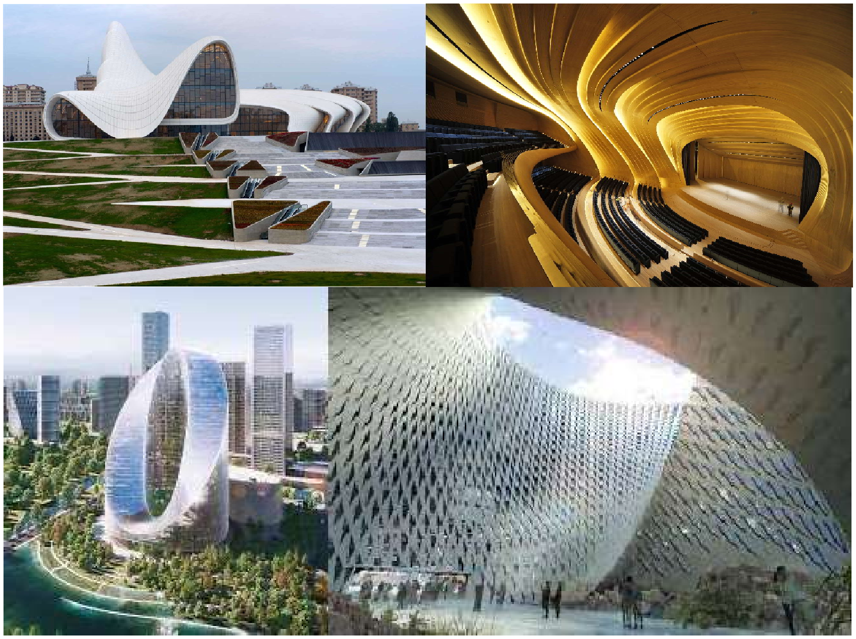

Fig. 2.10 : Examples of Parametricism - (*Top)* Views of Zaha Hadid's *Hyder Aliyev Center,* Baku*,* Azerbaijan, *(Bottom)* of Bjarke Ingels *Group's Infinity Loop,* Hangzhou, China

2.3 FUTURE VISION FOR CIVIL ENGINEERING

With demand for design and construction on the rise, the future holds many propositions. The key themes for civil engineering, at the juncture of **design + technology**, will be:

Sustainable Design & construction: Developing ***Green infrastructure*** that is safe, inclusive, energy efficient, and uses materials and techniques that have low environmental impact; using ***Nanotechnology and use of nano materials***, like, nano-silica, nano-clays, and iron nanoparticles, incorporated into concrete, and copper nanoparticles used in steel beams; and realising ***Futuristic transportation***, such as self-drive and travel pods through hyperloop will improve quality of life.

Cutting-edge technology: Using ***Augmented and Virtual reality*** to model real world construction challenges, as well as provide an immersive experience to visualise concepts via ***Digital Twins***; and leveraging ***AI, Big data and IoT*** to seamlessly harness data for mundane decisions and support collaborative projects with multiple stakeholders, will improve the quality and time of the design and construction.

The importance and profound impact of Civil engineering on society and the world, is undeniable and beckons that formidable and competent civil engineers forge the future.

UNIT SUMMARY

This Unit walks through an aspiring civil engineer on the importance of the profession, its remarkable lineage of breakthroughs across history and the future scope of the profession. Drawing from the knowledge shared in Unit 1, this Unit connects to the importance of the profession and practice as custodians of sustainable development and exemplifies the potential impact of civil engineering on achieving each SDG, either directly or indirectly. It further discusses the skills required for a competent civil engineer and the capabilities expected. The next Section elucidates marvels and breakthroughs in the ancient era up till present times, and continues to highlight in brief the major themes for the future.

EXERCISES

I. Multiple Choice Questions

Q. 2.1 Which of the following is not a Civil Engineering specialisation?
- (a) Geotechnical Engineering
- (b) Hydraulic Engineering
- (c) Environmental Engineering
- (d) Urban Planning

Q. 2.2 What are the SDGs that are directly impacted by Civil Engineering?
- (a) SDG 11 - Sustainable cities and communities,
- (b) SDG 6 - Clean water and sanitation
- (c) SDG 16 - Peace, justice and strong institutions
- (d) SDG 9 - Industry, innovation and infrastructure

Q. 2.3 Which ancient marvel, recognised as a UNESCO World Heritage site, is the oldest amongst the Seven Wonders of the World?
- (a) Three principal pyramids of Giza
- (b) Ziggurat with the 'White Temple' at Ur
- (c) Mohenjo-daro, Harappa and Lothal
- (d) The Stonehenge, United Kingdom

Q. 2.4 The Greeks and Roman style of architecture and construction is called?
(a) Renaissance
(b) Ancient
(c) Classical
(d) Gothic

Q. 2.5 Who is the Father of Renaissance?
(a) Leonardo da Vinci
(b) Andrea Palladio
(c) Leon Battista Alberti
(d) Filippo Brunelleschi

Answers of Multiple Choice Questions: 2.1 (d), 2.2 (c), 2.3 (a), 2.4 (c), 2.5 (d)

II. Short and Long Answer Type Questions

Q. 2.6 What are the roles and responsibilities of a Civil Engineer?

Q. 2.7 What were the major periods in design and construction? Write briefly on each with an example of civil engineering.

Q. 2.8 Describe the major future trends in Civil Engineering and elaborate which one is set out to be the most impact full in the days to come.